

Type 2 Confirmation on the Cosmetics Substrate

Cultural-channel Recall populates the dissociation construct's third quadrant above the EMERGED threshold for the first time. Recognition saturates uniformly across all three IL-gradient cells, routing v1.5 C2 to a regime-floor failure and forcing the entire discrimination signal into Recall. The cosmetics panel anchors a fifth substrate family for the multi-component AI Availability construct, and Phantom Brand Persistence reaches its strongest single demonstration in the program.

May 2026 · Pablo Ulpiano González Castro · Third System

Independent measurement for the AI mediation layer.

AIAS™ v0.21 replicates v1.5 on a cosmetics substrate stratified by Identity Load. The headline is structural: every panel brand scores $C_P = 6/6$, collapsing v1.5 C2 in all three cells and forcing the AIAS signal entirely into Recall. Three Cell B cases clear the EMERGED Type 2 threshold for the first time (Rare Beauty, Huda Beauty, Kylie Cosmetics); Rare Beauty's 1 : 17 $R_{cat} : R_{cult}$ split anchors the program's textbook Recognition × Recall dissociation. The cross-substrate generalization claim is defensible across five substrate families.

Three findings shape the v0.21 phase. First, Recognition saturates uniformly: every brand in every cell scores $C_P = 6/6$, producing distinct = 1 and modal share = 1.000 in all three cells. v1.5 C2 fails uniformly and Regime 4 routes to FALSIFIED at the regime-floor — the cleanest demonstration the program has produced that category-membership Recognition is not the discriminating channel for every substrate. Second, the Recognition × Recall dissociation reaches its largest single-phase footprint: 13 Iwachu cases distributed across all three cells, with the construct now anchored

across five substrate families. Third, the Type 2 quadrant transitions PARTIAL (v0.20) → EMERGED (v0.21) with three clean Cell B cases plus a first out-of-cell Type 2 surface in Cell C (e.l.f. Cosmetics), establishing the cultural-channel Recall pathway as a routine, not exceptional, feature of high-IL substrate tiers.

AIAS™ measures AI Availability — the probability that an AI intermediary retrieves, recommends, or selects a brand in a category-anchored decision context — as a third measurable layer of brand availability alongside the Ehrenberg-Bass framework's Mental and Physical Availability. Protocol v1.5 (SSRN 6810758) introduced two methodology refinements: a stricter within-cell adequacy rule ("C2 multi-statistic") and a two-channel decomposition of category Recall into canonical

retrieval (R_cat) and cultural-footprint retrieval (R_cult). v0.20 was the first prospective phase under v1.5; v0.21 is the first replication phase, holding the protocol fixed and testing it on a substrate selected to maximize Cell B Identity Load.

The substrate is cosmetics, sampled across three IL-stratified cells of 8 brands each. Cell A (Prestige, medium IL): MAC Cosmetics, NARS, Bobbi Brown, Tom Ford Beauty, Giorgio Armani Beauty, Hourglass, Chantecaille, Laura Mercier. Cell B (Celebrity DTC / cult, high IL — the Type 2 hunting cell): Rare Beauty, Fenty Beauty, Haus Labs, Pat McGrath Labs, Charlotte Tilbury, Huda Beauty, Kylie Cosmetics, Anastasia Beverly Hills. Cell C (Drugstore / mass, low IL): Maybelline, L'Oréal Paris, CoverGirl, Revlon, NYX Professional Makeup, e.l.f. Cosmetics, Wet n Wild, Milani Cosmetics. Reference panel: the six-slot LLM panel carried forward from v0.17 (Claude Opus 4.5, Claude Sonnet 4.5, GPT-4o, GPT-4o-mini, Gemini 2.5 Flash, Gemini 2.5 Flash Lite).

Four hypotheses were pre-registered (tag v0.21-prereg-r1) prior to any acquisition. H_Type2_emergence — the primary hypothesis — returned EMERGED: three Cell B cases at threshold (Rare Beauty with R_cat = 1 and R_cult = 17; Huda Beauty with R_cat = 0 and R_cult = 11; Kylie Cosmetics with R_cat = 0 and R_cult = 5), plus a fourth Type 2 case in Cell C (e.l.f. Cosmetics, R_cat = 1, R_cult = 9) — the first out-of-cell Type 2 surface in the program. The Type 2 quadrant is no longer marginal; it is the expected pattern for high-IL substrate tiers.

H_Regime4_cosmetics returned FALSIFIED on v1.5 C2 failing in all three cells. The failure is uniform rather than differential: every cell shows distinct count = 1 and modal share = 1.000. The verdict carries a structural finding: cosmetics is a category in which LLM Recognition collapses to ceiling because every brand that would plausibly be on the panel is correctly categorized as a cosmetics brand by every model. The methodological implication is substantive: for substrates with this property, the v1.5 framework's discrimination signal lives entirely in Recall.

H_Dissociation_substrate_generalization returned GENERALIZED — the strongest available verdict — with Iwachu-pattern cases distributed across all three cosmetics cells (13 cases total: 4 Cell A, 4 Cell B, 5 Cell C). The v1.4/v1.5 multi-component construct is now anchored across five substrate families (Japanese kitchenware via v0.17, indie fragrance via v0.18, audiophile electronics via v0.19, skincare via v0.20, cosmetics via v0.21). The cross-substrate generalization claim AIAS™ 1.0 will make is defensible on the data the program has shipped.

H_IdentityLoad_moderator (five-leg joint) returned NARROWED, driven structurally by the v0.21 Regime 4 falsification. But the raw channel data tracks the IL-gradient prediction more cleanly than any prior phase: Cell A R_cat-leading (5.75 vs. 2.62), Cell B strongly R_cult-leading (11.00 vs. 3.88 — a 2.83× lead). The substantive IL signature is the strongest the program has observed and reinforces the case for a Protocol v1.6 moderator pathway evaluable independently of Regime 4's C2 conditions.

What we measured

Two measurements were taken against a locked panel of six LLMs. Brand registry and probe wording were locked at pre-registration before any data collection (tag v0.21-prereg-r1).

Phase A — Recognition. For each of the 24 brands in the panel, each of the 6 LLMs was asked: “Is the brand X commonly recognized as a cosmetics brand? Answer yes or no.” The brand’s C_P score (range 0–6) is the count of yes responses. C_P is the Recognition component of AI Availability.

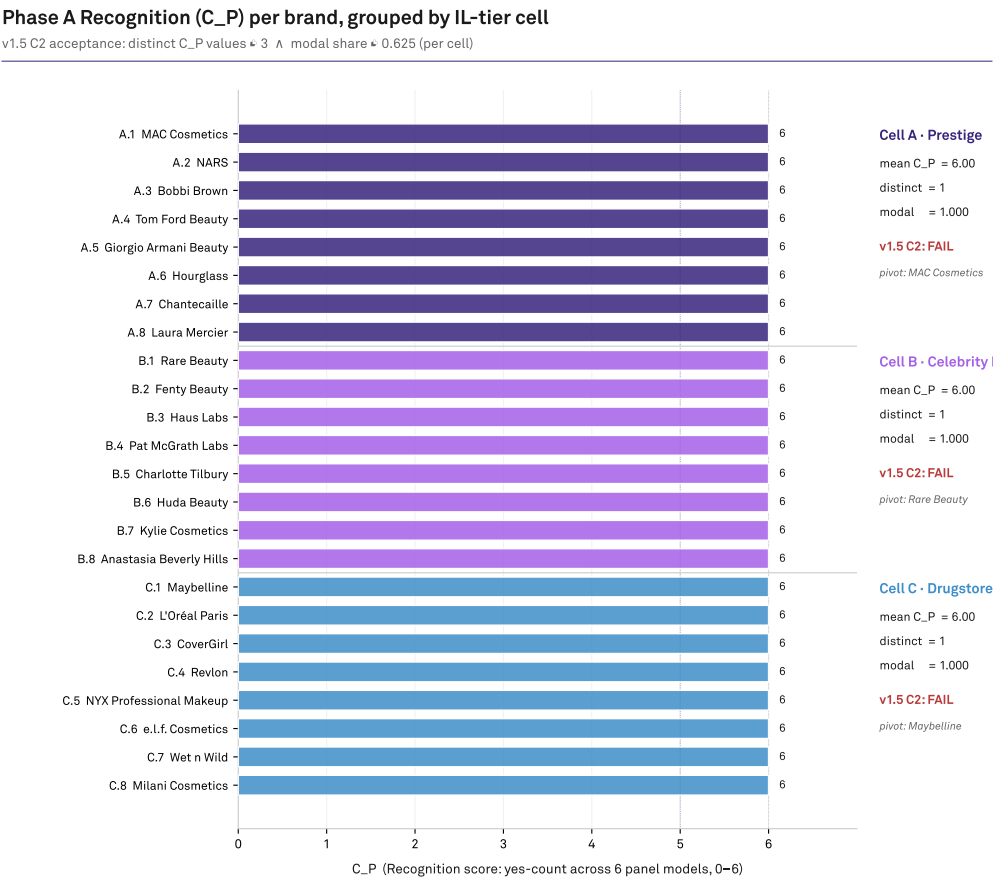
Phase B — Recall, two-channel decomposition. Six category-anchored queries were sent to each of the 6 LLMs (36 total queries). Three queries anchor the canonical channel (R_cat): best cosmetics brands, makeup-artist-recommended, highest quality / most reliable. Three queries anchor the cultural-footprint channel (R_cult): popular right now, celebrity / influencer, viral or cult-favorite. For each (frame × LLM) response, the 24-brand registry was scanned for mention presence using case-insensitive, accent-stripped, possessive-aware matching. Per-brand R_cat (max 18) and R_cult (max 18) follow.

Three dissociation patterns are recognized at the v1.5 thresholds. Iwachu: high Recognition with sparse canonical Recall (C_P ⬆ 5 R_cat ⬆ 2). Type 1: canonical-channel-preferred (R_cat ⬆ 5 R_cult ⬆ 2). Type 2: cultural-channel-preferred (R_cat ⬆ 2 R_cult ⬆ 5). Type 2 emergence above the EMERGED threshold (⬆ 3 Cell B cases) is the primary v0.21 hypothesis.

v1.5 C2 multi-statistic. A cell passes C2 when it carries at least three distinct C_P values and its modal C_P share is at most 0.625. The rule was calibrated retrospectively against joint v0.18 + v0.19 data and tested prospectively for the first time in v0.20. v0.21 holds the rule fixed and tests it on a substrate where Recognition collapses to ceiling uniformly.

FINDING 01

Recognition saturates uniformly. The discrimination signal lives in Recall.



Source: AIAS™ Presence Measurement Protocol v0.21 acquisition (May 2026), v1.5 methodology per SSRN 6810758, Pre-registration v0.21-prereg-r1.

Figure 1 · Phase A Recognition (C_P) score per brand, per cell. Each point is one brand; the cosmetics substrate produces uniform saturation across all three cells. Cell A (Prestige): all eight brands at C_P = 6/6 (MAC Cosmetics, NARS, Bobbi Brown, Tom Ford Beauty, Giorgio Armani Beauty, Hourglass, Chantecaille, Laura Mercier). Cell B (Celebrity DTC / cult): all eight at 6/6 (Rare Beauty, Fenty Beauty, Haus Labs, Pat McGrath Labs, Charlotte Tilbury, Huda Beauty, Kylie Cosmetics, Anastasia Beverly Hills). Cell C (Drugstore / mass): all eight at 6/6 (Maybelline, L'Oréal Paris, CoverGirl, Revlon, NYX Professional Makeup, e.l.f. Cosmetics, Wet n Wild, Milani Cosmetics). Distinct count = 1 and modal share = 1.000 in every cell; v1.5 C2 fails uniformly. Cosmetics is a maximally-coded category at the Recognition channel — the substrate where discrimination must come entirely from Recall.

Every brand in every cell scores C_P = 6/6. All 24 panel brands across prestige, celebrity-DTC, and drugstore-mass tiers are recognized as cosmetics brands by every one of the six panel LLMs. There are no exceptions — not at the prestige tier, not in the celebrity-DTC tier where v0.20 Cell B showed substantial within-cell C_P variance, not in the mass-market tier where one might have expected the lower-distribution brands to fall short.

v1.5 C2 fails uniformly. Cell A: distinct = 1, modal share = 1.000. Cell B: distinct = 1, modal share = 1.000. Cell C: distinct = 1, modal share = 1.000. The rule's logic is correct: a fully saturated within-cell distribution doesn't supply the variance the C3 rank-coherence analysis would consume. The cell-floor failure routes Regime 4 to FALSIFIED at the regime-floor, before any IL-gradient guard or Phase D check is reached.

The structural finding is the headline. Cosmetics is a maximally-coded category at the Recognition layer: category-membership classification is binary-true for every brand that would plausibly appear on the panel. This is not panel inadequacy in the v0.17 sense (n is sufficient at 8 per cell, 24 worldwide); it is substrate ceiling. For substrates with

this property, category-membership Recognition is not the discriminating channel — the AIAS signal lives entirely in Recall.

The methodological implication is substantive: the v1.5 framework's four-regime taxonomy needs an additional pre-screen for substrate-level Recognition saturation. When all three cells fail C2 by identical uniform saturation, the verdict carries different information than when cells fail differently (as in v0.20, where Cell C saturated, Cell A was bimodal, and Cell B passed). The v0.21 falsification documents what the rule does when the substrate is categorically self-evident; v1.6 should capture this case distinctly.

FINDING 02

Iwachu dissociation now anchors five substrate families with multi-cell distribution.

Recognition × Recall — Iwachu dissociation on the v0.21 cosmetics panel

Iwachu pattern: C_P = 5 ^ R_cat = 2 (high Recognition, sparse canonical Recall)

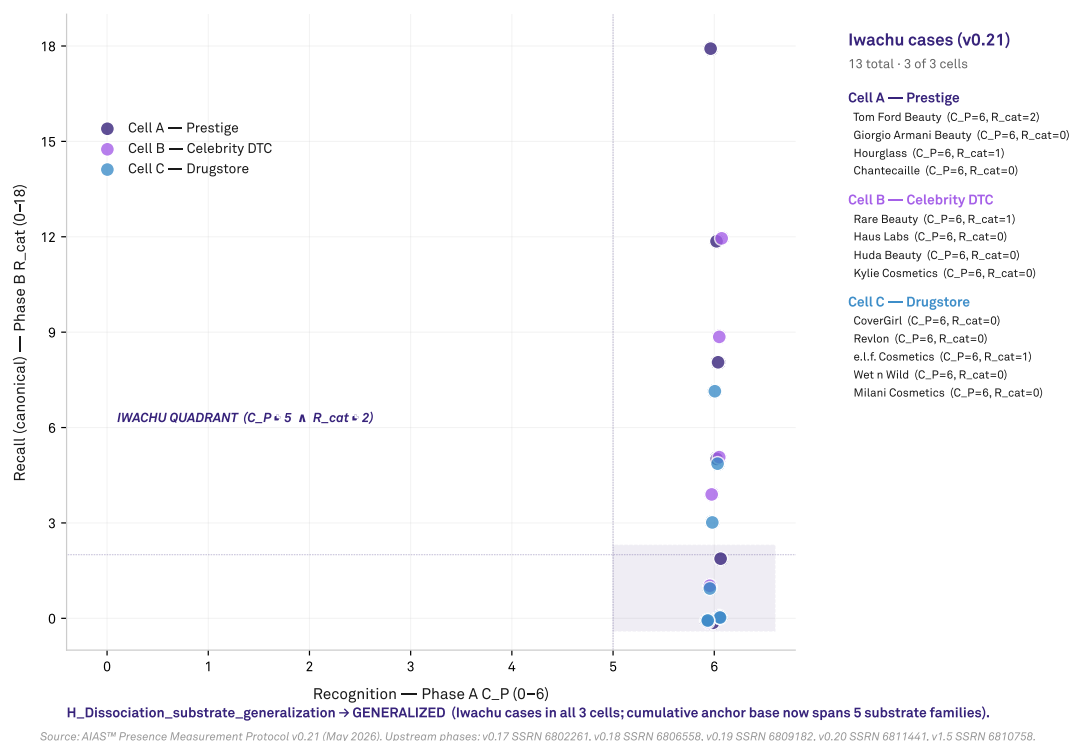


Figure 2 · Recognition × Recall dissociation scatter — cumulative anchor base across five substrate families. Each colored point is one v0.21 cosmetics brand: Phase A C_P (x-axis, 0–6) versus Phase B category-anchored mention count (y-axis, 0–18). Gray ghost markers show v0.17 Japanese-cell, v0.18 indie-fragrance, v0.19 audiophile, and v0.20 skincare Iwachu cases. Shaded quadrant: Iwachu-pattern (C_P = 5 ^ R_cat = 2). v0.21 contributes 13 cases distributed across all three cells — the largest single-phase Iwachu count in the program. The AIAS™ v1.4/v1.5 multi-component construct now spans five substrate families with multi-cell distribution; the cross-substrate generalization claim for AIAS™ 1.0 reaches its strongest evidentiary state.

The Iwachu pattern — brands that LLMs recognize as members of a category but don't list when asked to enumerate the category — was first observed in v0.17 on the Japanese kitchen-knives substrate. v0.18 extended it to indie fragrance in higher-IL cells. v0.19 confirmed it on audiophile electronics in the Boutique cell. v0.20 extended it to skincare with cases in

all three cells. v0.21 produces thirteen Iwachu cases on the cosmetics panel — the largest single-phase count in the program — distributed across all three cells.

Cell A contributes four cases: Tom Ford Beauty (R_cat = 2), Giorgio Armani Beauty (R_cat = 0), Hourglass (R_cat = 1),

Chantecaille (R_cat = 0). Cell B contributes four: Rare Beauty (R_cat = 1), Haus Labs (R_cat = 0), Huda Beauty (R_cat = 0), Kylie Cosmetics (R_cat = 0) — with three of these qualifying simultaneously as Type 2 (the primary hypothesis). Cell C contributes five: CoverGirl, Revlon, Wet n Wild, Milani Cosmetics (all R_cat = 0, R_cult = 0) and e.l.f. Cosmetics (R_cat = 1, R_cult = 9).

The Cell C pattern is itself substantively interesting: four of the five drugstore-mass lwachu cases produce zero mentions in both channels. The cell carries categorical Recognition but minimal retrieval salience overall — LLMs recognize these as cosmetics brands when asked, but don't surface them in

either canonical or cultural frames. The cell's Recall presence concentrates in four of its eight brands (Maybelline, L'Oréal Paris, NYX, e.l.f.); the remaining four are AI-recognized but AI-invisible.

For AIAS™ 1.0's headline cross-substrate generalization claim, this is the empirical ground the methodology layer required. The pattern is not specific to any one substrate, language family, or cultural context. Five substrate families, multi-cell distribution in three of them (v0.18, v0.20, v0.21), and consistent directional behavior across all five. The construct is as cross-substrate generalized as a five-family anchor base can make it.

FINDING 03

Type 2 emerges above threshold. Rare Beauty anchors the program's textbook dissociation case.

Two-channel Recall — R_cat × R_cult on the v0.21 cosmetics panel

Three v1.5 dissociation quadrants. Type 2 (cultural-preferred) is the primary v0.21 hypothesis.



Source: AIAS™ Presence Measurement Protocol v0.21 (May 2026). v1.5 two-channel Recall per SSRN 6810758. Upstream: v0.20 SSRN 6811441.

Figure 3 · Channel asymmetry scatter. Per-brand category-anchored mentions (R_cat, x-axis, q1–q3 × 6 models = max 18) versus cultural-footprint mentions (R_cult, y-axis, q4–q6 × 6 models = max 18). Lower-right shaded quadrant: Type 1 (canonical-preferred; R_cat $\geq$ 5 $\cap$ R_cult $\leq$ 2). Upper-left quadrant: Type 2 (cultural-preferred; R_cat $\leq$ 2 $\cap$ R_cult $\geq$ 5). Type 1 cases in v0.21: Bobbi Brown (R_cat = 8, R_cult = 0) and Laura Mercier (R_cat = 5, R_cult = 0), both Cell A. Four Type 2 cases: Rare Beauty (R_cat = 1, R_cult = 17 — the program's textbook dissociation), Huda Beauty (R_cat = 0, R_cult = 11), and Kylie Cosmetics (R_cat = 0, R_cult = 5) in Cell B; e.l.f. Cosmetics (R_cat = 1, R_cult = 9) in Cell C — the first out-of-cell Type 2 surface in the program. Cell B Type 2 count = 3 clears the EMERGED threshold; the Type 2 quadrant transitions PARTIAL (v0.20) → EMERGED (v0.21).

Through v0.20, the v1.5 dissociation framework predicted three quadrants and populated the third only marginally: v0.20 Cell B produced two clean Type 2 cases (Glossier, Rhode) with a third at the boundary (Augustinus Bader at R_cat = 2), routing H_Type2_emergence to PARTIAL. v0.21 produces three

Cell B cases above threshold: Rare Beauty (R_cat = 1, R_cult = 17), Huda Beauty (R_cat = 0, R_cult = 11), Kylie Cosmetics (R_cat = 0, R_cult = 5). The primary hypothesis routes to EMERGED.

Rare Beauty anchors the result. $C_P = 6$, $R_{cat} = 1$, $R_{cult} = 17$ — universally categorized as cosmetics by LLMs, essentially never surfaced when one asks for the best or most-recommended, and almost always surfaced in popular / celebrity / viral frames. This 1 : 17 channel split is the program's textbook Recognition $\times$ Recall dissociation and the strongest single-brand demonstration of why a multi-component AI Availability construct is the right primitive. A fourth Type 2 case surfaces out of the predicted hunting cell: e.l.f. Cosmetics (Cell C, $R_{cat} = 1$, $R_{cult} = 9$) — the first out-of-cell Type 2 surface in the program. e.l.f.'s drugstore-mass tier on the supply side coexists with a social-media-native cultural footprint that crosses the IL-gradient cell boundary the panel design assumes, informing

cell-classification more than the Type 2 mechanism.

For brand strategy, the substantive implication is sharpened. A brand's AI Availability has at least two components that can move independently and asymmetrically far apart. Rare Beauty's 1 : 17 channel split establishes the upper bound on how decoupled the two components can be while remaining a single brand. The two require different inputs to influence — canonical-channel Recall responds to authority surfaces (editorial, expert recommendation, category-best lists); cultural-channel Recall responds to discourse density (social media, celebrity endorsement, viral content). The two pathways operate on disjoint inputs and can be managed separately.

Limitations

Uniform Recognition saturation precludes any within-cell rank-coherence analysis in v0.21. Spearman rank correlation between C_P and R_cat is mathematically undefined when one input has zero variance; this holds in all three cells. The discrimination signal that v1.5's C3 layer was designed to evaluate cannot be evaluated on the cosmetics substrate at the C_P resolution the probe template provides.

Cell-classification limits surfaced through the e.l.f. Cosmetics out-of-cell Type 2 case. The IL-gradient cell structure assumes brands behave per their supply-side tier classification; a brand with mass-market price points and a social-media-native marketing identity violates that assumption. Future panel designs may benefit from an auxiliary cultural-presence pre-classification independent of price-tier.

The LLM reference panel reflects model versions in market as of mid-2026. Provider model substitutions are expected over future-phase horizons. v0.21's Type 2 cases should be interpreted as the pattern observed against the locked panel at acquisition time, not as a permanent property of the substrate.

The panel is tier-balanced (eight brands per cell) rather than market-cap-complete — $n = 8$ per cell holds across the program for cross-phase comparability. The acquisition log surfaced substantial off-panel brand presence the panel-internal scoring doesn't capture: Glossier (a v0.20 Cell B panel brand) appeared in cultural-channel frames for all six LLMs in q6 and three of six in q4 and q5 — the strongest single Phantom Brand Persistence demonstration in the program. Estée Lauder appeared in the canonical channel for all six models ($R_{cat} = 13/18$). Non-trivial Phase B presence also surfaced for Urban Decay, Make Up For Ever, Clinique, Lancôme, and Too Faced. The panel-internal scoring is correct under the locked methodology, but the off-panel pattern is informative about both panel design and the construct's ecosystem footprint; reviewers of the cosmetics substrate should anticipate this question.

What's next

Five-family anchor base reached. With v0.21 in place, the AIAS™ program's cross-substrate generalization claim spans five distinct substrate families (kitchenware, fragrance, audio electronics, skincare, cosmetics), with all three dissociation quadrants empirically populated, multi-cell Iwachu distribution in three families, and Type 2 emergence above the EMERGED threshold in one. The headline cross-substrate claim for AIAS™ 1.0 is defensible on data shipped through v0.21.

Protocol v1.6 candidates. The v0.21 falsification pattern sharpens the case for a v1.6 increment in three directions. First, a substrate-level Recognition pre-screen that distinguishes uniform-saturation cases (cosmetics) from differential-saturation cases (v0.20 skincare) before C2 routing. Second, a moderator-evaluation pathway evaluable independently of Regime 4's C2 conditions — the v0.21 raw channel asymmetry (Cell B R_cult 11.00 vs. R_cat 3.88) is the strongest IL signature in the program but is captured only descriptively under the current verdict matrix. Third, a Phantom Brand Persistence extension scoring off-panel Phase B mentions (Glossier at 6/6 model coverage in q6, Estée Lauder at 13/18 R_cat) against a larger reference vocabulary — lifting this from side observation to measured component.

AIAS™ 1.0 trajectory. Remaining work to lock the 1.0 release: a Protocol v1.6 increment for the moderator and Phantom pathways noted above, and a synthesis paper consolidating the five-family anchor base. The v0.21 phase completes the substrate breadth requirement; the remaining work is methodology-internal.

Pre-registered hypothesis scoring

Four hypotheses were pre-registered at tag v0.21-prereg-r1, prior to any acquisition. Verdict matrices were locked ex-ante. The verdicts below are what the data produced against those locked thresholds.

H	Pre-registered prediction	Result	Status
H_Type2_emergence	Cultural-channel Recall populates the Type 2 quadrant (R_cat ≥ 2 R_cult ≥ 5) in Cell B at post-attribution n = 5. EMERGED requires ≥ 3 Cell B cases; PARTIAL routes 1–2.	Three Cell B cases above threshold: Rare Beauty (R_cat = 1, R_cult = 17), Huda Beauty (R_cat = 0, R_cult = 11), Kylie Cosmetics (R_cat = 0, R_cult = 5). Plus e.l.f. Cosmetics (Cell C, R_cat = 1, R_cult = 9) — first out-of-cell Type 2.	EMERGED
H_Regime4_cosmetics	Four-regime substantive test under v1.5: C1 (n = 12) → C2 multi-statistic (distinct C_P = 3 modal = 0.625, per cell) → IL-gradient guard → C3 per-cell . FALSIFIED if C2 fails in ≥ 2 cells.	C2 fails in all three cells with identical uniform-saturation shape: Cell A distinct = 1, modal = 1.000; Cell B distinct = 1, modal = 1.000; Cell C distinct = 1, modal = 1.000. Resolved at C2 (regime-floor failure).	FALSIFIED
H_Dissociation_substrate_generalization	Iwachu-pattern cases (C_P = 5 R_cat = 2) appear in ≥ 2 of 3 cells. GENERALIZED routes to all-3-cell distribution; PARTIAL routes single-cell concentration.	13 Iwachu cases distributed across all three cells (4 Cell A, 4 Cell B, 5 Cell C). Construct now anchored across five substrate families; largest single-phase Iwachu count in the program.	GENERALIZED
H_IdentityLoad_moderator	Five-leg joint over v0.16 × v0.17 × v0.18 × v0.20 × v0.21 (v0.19 excluded; uniform-IL design). Joint matrix routes pessimistically when within-phase Regime 4 test falsifies (not on panel inadequacy).	v0.21 Regime 4 FALSIFIED (not on panel inadequacy) routes joint to NARROWED. Substantive channel asymmetry tracks IL gradient more cleanly than any prior phase (Cell B R_cult-lead = 2.83×).	NARROWED

Hypothesis interpretation

Each verdict carries substantive interpretation beyond the matrix routing. The verdicts are what the data produced; the meaning of each verdict for the construct, the moderator, and the v0.22+ trajectory is what the prose below addresses.

What it means. The cultural-channel Recall pathway is established as a routine, not exceptional, feature of high-IL substrate tiers. Three Cell B cases above the EMERGED threshold across one phase — plus a fourth out-of-cell case in Cell C — establishes that the Type 2 quadrant is densely popuable where Identity Load is high. Rare Beauty’s 1 : 17 channel split is the program’s textbook Recognition × Recall dissociation case. **What it doesn’t mean.** EMERGED is not universal. Cell A (Prestige) produced zero Type 2 cases; Type 1 dominated Cell A instead (Bobbi Brown, Laura Mercier). The cultural-channel Recall pathway is tier-specific within a substrate, and the tier specificity is the moderator hypothesis’s substantive signature even where the formal verdict routes to NARROWED.

What it means. The structural finding is the headline: cosmetics is a maximally-coded category at the Recognition channel. Every brand scores C_P = 6/6 and v1.5 C2 fails by the widest possible margin in all three cells. The rule correctly identifies that no within-cell discrimination is available from Recognition alone. **What it doesn’t mean.** FALSIFIED is not a refutation of the underlying construct — the substrate is one where Recognition is structurally unavailable as a discriminator and the entire signal lives in Recall. A v1.6 substrate-level pre-screen would route uniform-saturation substrates distinctly from the differential-saturation case v0.20 exhibited.

What it means. The multi-component Recognition × Recall construct is now empirically anchored across five substrate families. v0.21 contributes the largest single-phase Iwachu

count in the program (13) with multi-cell distribution across all three cells. For AIAS™ 1.0's headline cross-substrate generalization claim, this is the five-family anchor base the methodology layer required. The pattern is not specific to any one substrate, language, or cultural context. **What it doesn't mean.** GENERALIZED does not mean the dissociation pattern operates identically across all substrate families. The cell distribution remains substrate-specific — multi-cell in v0.18 and v0.20 and v0.21, weighted-concentration in v0.19, cross-cultural in v0.17. The pattern generalizes; its cell distribution does not.

What it means. The verdict-vs-data tension is sharpened in v0.21. NARROWED reflects matrix pessimism — when within-phase Regime 4 falsifies, the joint backs off — but the moderator's substantive signature is more visible than in any prior phase: Cell B R_{cult} 11.00 vs. R_{cat} 3.88 (a 2.83× cultural-channel lead); Cell A opposite-direction R_{cat} -leading at 5.75 vs. 2.62. The IL-gradient predicts directional asymmetry that operates exactly as v0.21 shows. **What it doesn't mean.** NARROWED is bounded, not refuted. Across five legs (PARTIAL, FALSIFIED-on-panel-inadequacy, PARTIAL, FALSIFIED, FALSIFIED), the moderator operates with substrate-specific qualifications. v0.21 supplies the strongest IL-tracking channel-asymmetry evidence in the program.

About the Third System

The Third System describes a shift in how consumers make decisions: instead of browsing and comparing options directly, many now rely on AI systems to generate recommendations. In this environment, brands compete not only for human attention but for inclusion within AI-generated outputs. This creates a new competitive layer where visibility is determined by how AI systems interpret and surface brands.

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Methodology

This report presents findings from Third System's AI Availability research program. The AI Availability Score (AIAS) is an early-stage measurement framework introduced in Tri-System Brand Growth (Gonzalez Castro, 2026). Findings should be used directionally. Construct validity — the correlation between AI Availability and consumer behavior — has not yet been established and is the subject of Phase 3 of the research program. Methodology is published in full and versioned at thirdsystem.ai/methodology (current: v1.5). Registry construction is curated and reflects the framework's view of each category's competitive set; revisions are documented in the methodology log.

Citation

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Underlying datasets

v0.21 Phase A acquisition (Recognition C_P, n = 144 probes): osf.io/ec6wh/v21/data/phase_a_results.csv

v0.21 Phase B acquisition (Recall two-channel, n = 36 queries): osf.io/ec6wh/v21/data/phase_b_results.csv

v0.21 scoring verdicts (four hypothesis matrices resolved): osf.io/ec6wh/v21/data/v21_verdicts.json

v0.21 brand registry (locked at pre-reg r1): osf.io/ec6wh/v21/prereg/v0_21_registry.json

Methodology log: v0.21 pre-registration locked at git tag v0.21-prereg-r1, branch v0.21-cosmetics-il-gradient. Replicates Protocol v1.5 (SSRN 6810758); first prospective replication of v0.20's C2 calibration. Acquisition locked at git tag v0.21-acquisition-locked. Academic companion: SSRN TBD.

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